

Shreesha G. Bhat

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RESEARCH INTERESTS

Distributed Systems, Data Streaming, Systems for Agentic AI, Machine Learning Systems, Storage Systems, CXL

EDUCATION

University of Illinois Urbana-Champaign 2023 - 2028 (expected)
PhD in Computer Science | Advisors: [Ram Alagappan](#), [Aishwarya Ganesan](#) | CGPA: 4.0/4.0

Indian Institute of Technology, Madras 2018 - 2023
Dual Degree (BTech + MTech) in Computer Science & Engineering | Advisor: [Kartik Nagar](#) | CGPA: 9.66/10.00

PUBLICATIONS

- OSDI '26** MEGALON: Efficient Data Sharing for Partly Coherent CXL Memory
Jiyu Hu, Seokjoo Cho, Landon Johnson*, Kiran Hombal, Shreesha G. Bhat, Marcos K. Aguilera, Ramnatthan Alagappan, Aishwarya Ganesan (*equal contribution)*
The 20th USENIX Symposium on Operating Systems Design and Implementation ([OSDI](#)) 2026
- EuroSys '26** A Logically Disaggregated Cache for Replicated Storage Systems
Kiran Hombal, Henry Zhu, Shreesha G. Bhat, Neil Kaushikkar, Ramnatthan Alagappan, Aishwarya Ganesan
The 21st European Conference on Computer Systems ([EuroSys](#)) 2026
- OSDI '25** Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering
Shreesha G. Bhat, Tony Hong, Xuhao Luo, Jiyu Hu, Aishwarya Ganesan, Ramnatthan Alagappan
The 19th USENIX Symposium on Operating Systems Design and Implementation ([OSDI](#)) 2025
Covered on the [Metadata Blog](#)
- SOSP '24** LazyLog: A New Shared Log Abstraction for Low-Latency Applications
Xuhao Luo, Shreesha G. Bhat, Jiyu Hu*, Ramnatthan Alagappan, Aishwarya Ganesan (*equal contribution)*
The 30th Symposium on Operating Systems Principles ([SOSP](#)) 2024
Best Paper Award | Invited to [Transactions on Computer Systems](#) | Inspired [Lightning Topics](#) in Warpstream
- FMCAD '23** Automating Cutoff-based Verification of Distributed Protocols
Shreesha G. Bhat, Kartik Nagar
Formal Methods in Computer-Aided Design ([FMCAD](#)) 2023
- DISC '21** Brief Announcement: Automating and Mechanising Cutoff Proofs for Parameterized Verification of Distributed Protocols
Shreesha G. Bhat, Kartik Nagar
35th International Symposium on Distributed Computing ([DISC](#)) 2021

REFEREED JOURNAL PUBLICATIONS

- TOCS '25** LazyLog: A New Shared Log Abstraction and Design for Modern Low-Latency Applications
Xuhao Luo, Shreesha G. Bhat, Jiyu Hu*, Ramnatthan Alagappan, Aishwarya Ganesan (*equal contribution)*
ACM Transactions on Computer Systems ([TOCS](#)) August 2025
Invited for Publication

RESEARCH EXPERIENCE

Research @ Distributed and Storage Systems Lab | UIUC Aug '23 – present
Research Assistant | Guide: [Ram Alagappan](#), [Aishwarya Ganesan](#)

- Designed and implemented multiple shared log abstractions and systems for modern application paradigms.
- Contributed to projects on CXL data sharing and replicated storage systems.

Networking and Distributed Systems Lab | Hewlett Packard Labs May'25 – August'25
Research Associate Intern | Guide: [Puneet Sharma](#)

- Designing infrastructure for agentic AI applications.

Improving Cloud Reliability through Systematic Testing | Microsoft Research India Aug '22 – Jul '23
Research Intern | Guide: [Akash Lal](#)

- Worked on improving reliability of Azure Cloud Services using concurrency testing tools such as [Coyote](#) for C++ programs.

- Built a deterministic concurrency testing framework for a production scale replication library, Azure RSL, which provides an implementation of the Paxos consensus algorithm. Implemented several optimizations to improve state-space coverage.

Parameterized Verification of Distributed Protocols | IIT Madras

Sep '20 – Jul '23

Young Research Fellow | Guide: Kartik Nagar

- Investigated cutoff-based techniques for verifying that distributed protocols meet their specification irrespective of the size of the parameter they are instantiated with (such as number of nodes).
- Proposed a framework to mechanize simulation based proofs for cutoffs and applied the approach on a variety of distributed protocols using Z3 as a backend SMT solver.

AWARDS & ACHIEVEMENTS

- Awarded the **Best Paper Award** at SOSP 2024 for the paper titled *LazyLog: A New Shared Log Abstraction for Low-Latency Applications*.
- Selected as a undergraduate research fellow as part of the (YRF) program.
- Secured prizes for excellent academic performance in the 1st, 2nd, 7th and 8th semesters at IIT Madras.
- Secured All India Rank of **851** and **619** in JEE (Joint Entrance Examination) Advanced & Mains 2018.
- Qualified for KVPY fellowship with an All India Rank of **142**.
- Among the **top 300** in India qualified to write national olympiad examinations for Physics, Chemistry and Astronomy (INPhO, INChO, INAO).

SERVICE

- **Artifact Evaluation Committees:** OSDI'24, ATC'24, FAST'26 (Spring and Fall)
- **Shadow Program Committees:** EuroSys'26
- **Student Volunteer:** SOSP'24

TEACHING

- **CS598 Storage Systems, TA, Fall 2025:** Held office hours, graded assignments, and led lecture sessions.

GRANTS

- **Conference Travel Grants:** OSDI'25, SOSP'24